

Dial Tool

Guidebook

beta 2.3 · 4 Oct 2026

A watch-dial design tool for DrawBot on macOS

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Start

Dial Tool is one Python file that runs inside DrawBot, a free drawing app for macOS. It draws a dial as a list of rings, three hands and a date window, in millimetres.

Install

Three steps, one file.

- 1 **DrawBot 3.132** or newer, from drawbot.com.
- 2 **dial.py** from github.com/duztman/dial-tool. Branch main is the released version; dev is the newest work.
- 3 Open dial.py in DrawBot and press **⌘R**. The window opens.

Speed

One optional package.

Knockouts and hollow shapes are cut by combining outlines. DrawBot's own routine is slow; skia-pathops is 2 to 14 times faster.

In DrawBot: Python → Install Python Packages → **skia-pathops** → Go. The Export log then starts with “shape combining: skia-pathops”.

Memory

Nothing to save.

Settings are stored as you work. Pressing **⌘R** again, or quitting DrawBot, keeps them: the last session reopens.

Export → Save... writes a named copy; Load... opens one. The nine dials in this guide come as such files.

Unit

mm

Every length is in millimetres, drawn at true size. Radii are measured from the dial's centre.

Angle

0° = 12

Clock angles, growing clockwise. 90° is 3 o'clock, 180° is 6, -90° is 9.

Shape

Fills

Everything exported is a filled outline, never a stroke. A 0.12 mm tick is exactly 0.12 mm in Illustrator.

Order

Top first

The ring list is the stacking order. The top ring is drawn last, in front of the others.

The window

Settings on the left, the dial on the right, time underneath. The next page shows it whole, redrawn in grey from the tool's own measurements, with the field watch loaded.

1

Sections

Dial, Rings, Hands, Date, Export. One is shown at a time.

2

Ring list

Top ring in front. The checkbox hides a ring; double-click a name to change it. The selected ring is the one the settings edit.

3

Ring tools

Add ring... puts a new ring above the selected one. Duplicate, Delete, move up, move down.

4

The ring's settings

One scrolling list. Rows that don't apply to the ring's kind are hidden, not greyed out.

5

Preview

Follows every change. Pinch or ⌘-scroll zooms about the cursor; two fingers pan. With Guides on, the selected ring's edges are orange.

6

Time

Drag, or type 10:09:36. Now sets the clock's time; Play runs the hands.

7

Guides, Canvas · PDF, Fit

Guides are never exported. Canvas is the fast preview; PDF is DrawBot's own viewer, the reference. Fit shows the whole page.

8

Status

Ring, font, diameter; then how long the geometry (build) and the drawing (frame) took.

Number

Slider, field and stepper show one value.

Length

Width · Ø

Drag for a rough value, type for an exact one, step for the last hundredth. Typing past a slider's end extends the slider.

Typing

Fields do arithmetic.

1.2	sets 1.2
+0.5	adds 0.5
x0.5	halves
/3	a third

Choice

Popups, switches, checkboxes.

Kind

Shape

Style

☐ Skip those instead

A choice can change which rows are shown: pick dot and Length disappears.

Dial Rings Hands Date Export

Ring	Kind
☒ Hours	numerals
☒ 24 hours	numerals
☒ Hour marks	markers
☒ Minutes	ticks

Add ring... Duplicate Delete ↑ Up ↓ Down

Kind ticks

Colour

Outer radius 14.40

Positions

Count 60

Start ° 0.0

Span ° 360

Every 1

Offset 0

☐ Skip those instead

Give way to rings above

Skip where markers above

Knock out nothing

Look

Shape bar wedge dot file

Length 1.00

Width · Ø 0.16

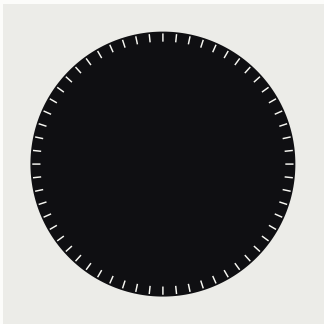
Round outer 0.00

Round inner 0.00

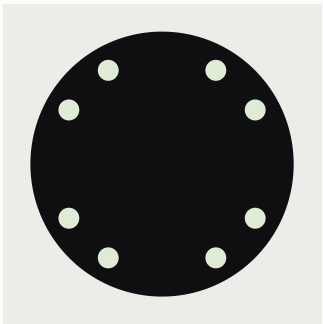


Rings

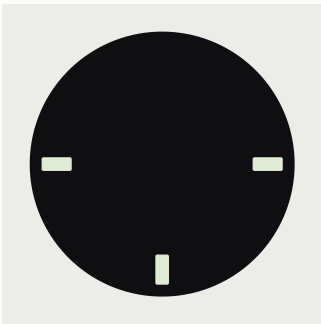
A ring repeats one thing round the centre, at its positions. A dial is a short list of rings, stacked.



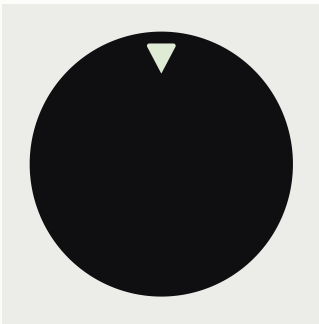
Minutes
ticks · Count 60



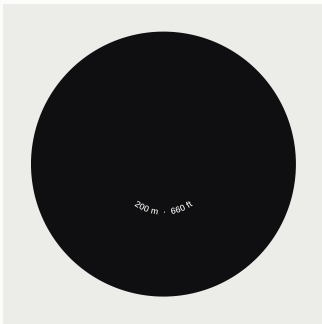
Dots
markers · Every 3, skip those instead



Bars 6 · 9
markers · Count 4 · 12 o'clock: none



Triangle
markers · Count 1 · wedge



Depth
numerals · Custom · on path



The dial
with hands and the date

The idea

One dial, five rings.

The diver on page 16, taken apart. Each ring does one job and the list stacks them. Nothing is special-cased: the triangle is a markers ring with a count of one, the depth rating a numerals ring with one label.

The list

Ring	Kind
☒ Triangle	markers
☒ Bars 6 · 9	markers
☒ Dots	markers
☒ Depth	numerals
☒ Minutes	ticks

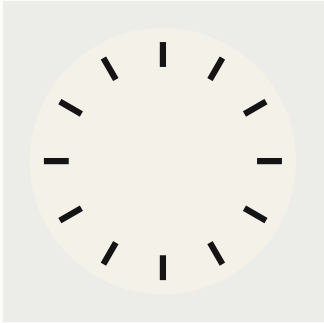
Kind

Four kinds, changeable at any time.

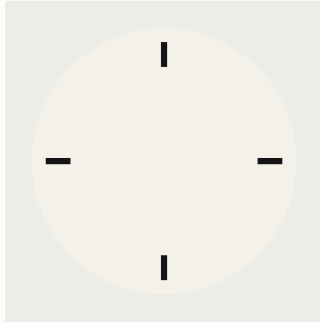
Ticks and **markers** repeat a shape. **Numerals** repeat a label. A **band** is colour between two radii. Every ring carries every setting, so changing a ring's kind loses nothing.

Positions

Where a ring puts things. Six numbers describe almost any track, scale or set of hours.



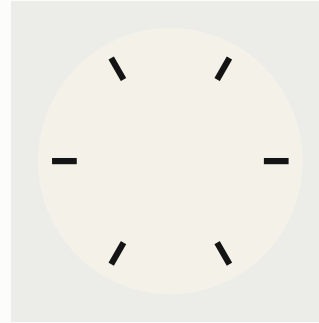
Count 12
Start 0° · Span 360°



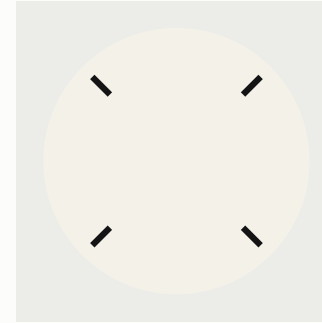
Every 3
the cardinals



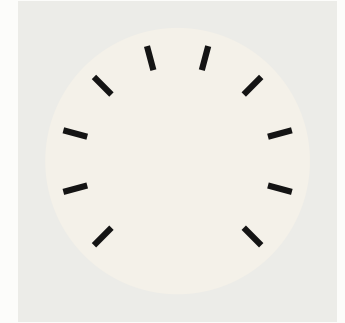
Every 3 · skip those instead
the other eight



Every 2 · Offset 1
the odd hours



Count 4 · Start 45°
a quarter turn off



Count 10 · Start -135° · Span 270°
an arc: both ends marked

Count · Start · Span

A full turn spaces evenly. An arc marks both ends.

Count

Start °

Span °

With Span 360° twelve positions sit 30° apart. With less, the first and last sit on the ends of the arc: Count 10 over 270° is a scale from 0 to 9.

Every · Offset · Skip those instead

Use some of the positions.

Every

Offset

☒ Skip those instead

Every 3 keeps the cardinals. Tick Skip those instead and the same setting keeps the other eight. Offset moves the starting point.

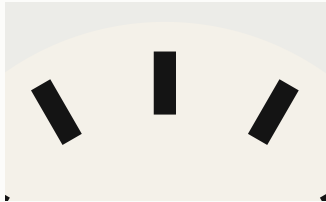
Radius

Measured to the outer edge.

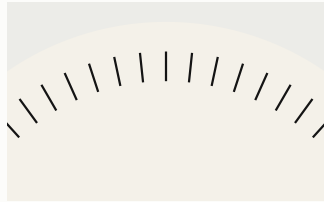
For ticks, markers and bands, Radius is the outer end; Length or Band width runs inward from it. For numerals, Radius is the centre of the text.

Marks

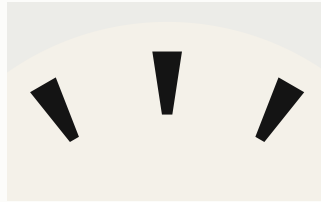
Ticks and markers share one set of shapes. A markers ring adds a first position that can differ from the rest.



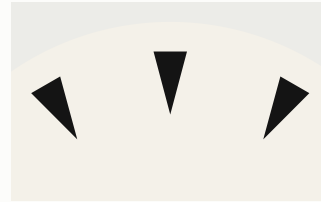
bar
Length 3.4 · Width 1.2



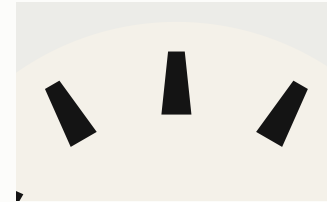
bar, fine
Count 60 · Length 1.6 · Width 0.12



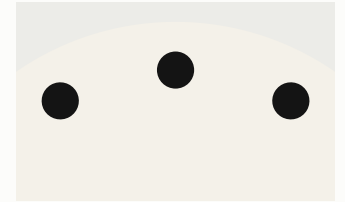
wedge
Inner width × 0.35



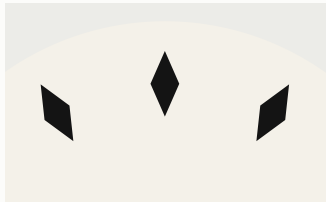
wedge
Inner width × 0



wedge
Inner width × 1.8



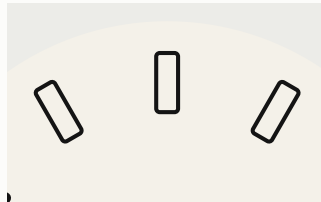
dot
Ø 2.0



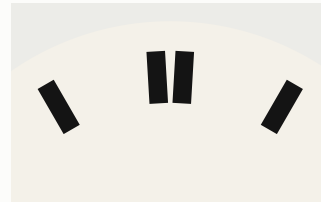
file
your own SVG · PDF · AI



rounded
Round outer 0.6 · Round inner 0.2



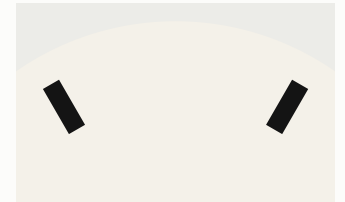
hollow
Hollow wall 0.22



12: double
Gap 0.4



12: triangle
Scale × 1.2



12: none

Shape

Bar, wedge, dot, or your own file.

Shape

bar

wedge

dot

file

Length

3.40

Width · Ø

1.20

Round outer

0.60

Round inner

0.20

Hollow wall

0.00

Wedge and corners

One number tapers; two round.

Inner width × sets the inner end as a share of the outer: 0 is a triangle, 1 a bar, more than 1 widens inward.

Round outer and Round inner are true fillets, in mm. Hollow wall leaves an outline of that thickness.

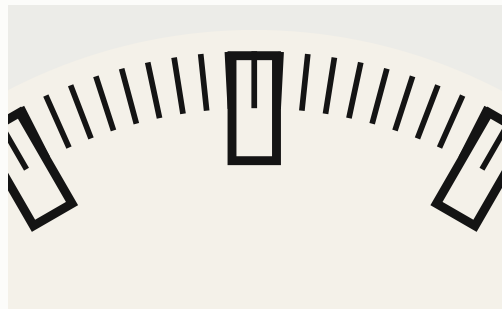
12 o'clock

The first position can differ.

A markers ring draws its first position the same, double (with a Gap), as a triangle, or not at all. Scale × sizes it.

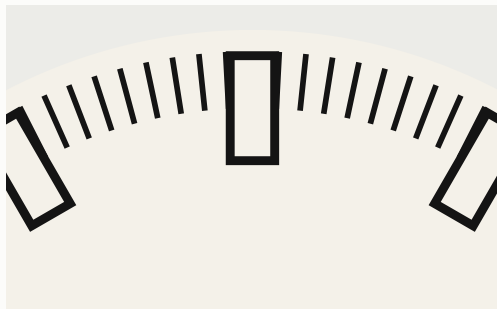
Give way

How a ring reacts to the rings above it. One mechanism covers every pair of rings.



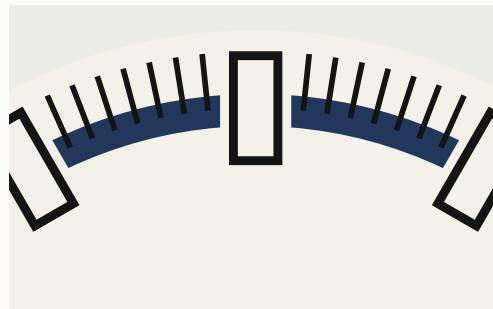
stack

Skip where: nothing · Knock out: nothing



skip where

markers above



knock out

markers above · Clearance 0.25



knock out

markers & numerals above

Skip where

Leave out positions that are taken.

The ring drops its own positions wherever the chosen rings above already have a mark: minute ticks that stop for the hour markers, fine ticks that stop for the minute ticks.

Knock out

Cut a gap round what is above.

The ring is cut back by Clearance mm round the shapes above it: markers, markers and numerals, or everything. The date window always clears every ring, by its own Print clearance.

The ticks in the third picture

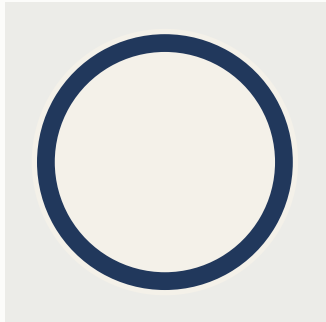
Skip where

Knock out

Clearance

Bands

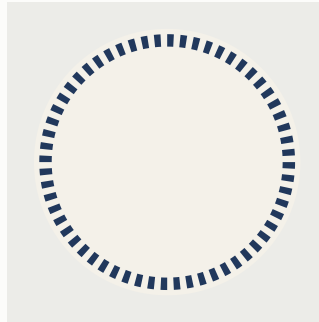
Flat colour between Radius and Radius – Band width: rings, rails, zones and sectors.



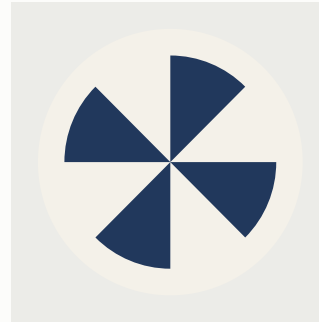
ring
Count 1 · Band width 2.0



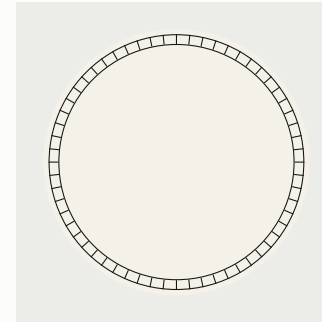
sectors
Count 12 · Every 2



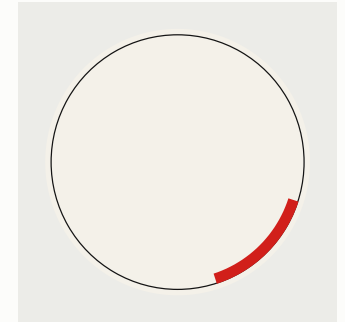
fill
Count 60 · Fill 50 %



pies
Band width = radius



rails
two bands, Band width 0.12



arc
Start 108° · Span 54°

Segments

Positions cut a band into pieces.

Count 1 is a full ring. More positions cut it into segments that tile the span; Fill % is how much of its step each segment covers.

Uses

Rails, zones, sectors.

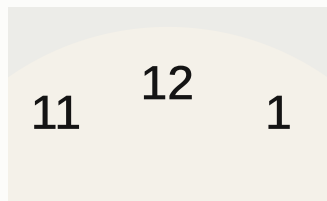
Railroad rails are two bands 0.12 wide. A band as wide as its radius reaches the centre: with Every 2 that is a sector dial. A short span is a red zone.

The sectors in the second picture

Outer radius	14.40	↑ ↓
Band width	3.00	↑ ↓
Count	12	↑ ↓
Every	2	↑ ↓
Fill % of step	100	↑ ↓

Numerals

One label per position: from a built-in set, a count, or a list of your own.



Arabic



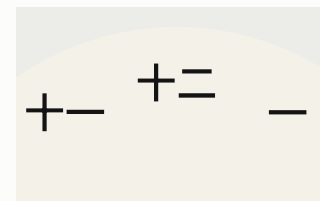
Roman · III



Roman · IV



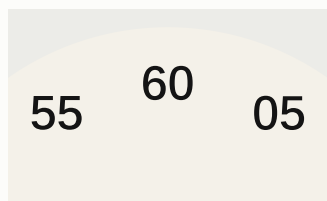
Eastern Arabic



Chinese

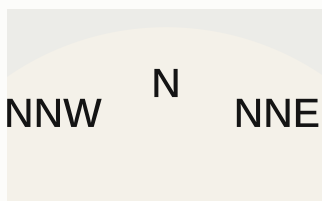


Words
on path



Numbers

First 5 · Step 5 · Digits 2 · Start 30°



Custom

N, NNE, , , E, ...

Numbers

Labels that count.

Labels

First number

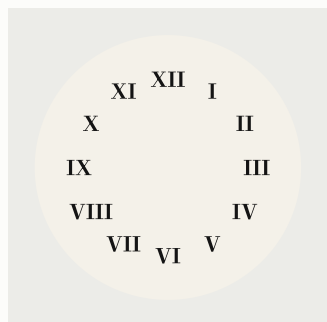
Step

Digits

Custom

Your own list, comma-separated.

One entry per position. An empty entry leaves that position empty, which is how two rings can share one circle of hours.



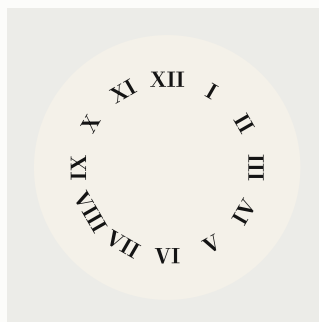
upright

always level



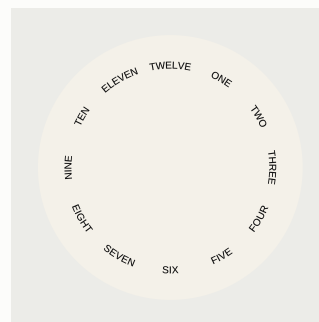
radial

turns with the circle



radial, auto-flip

never upside down



on path

letters follow the circle

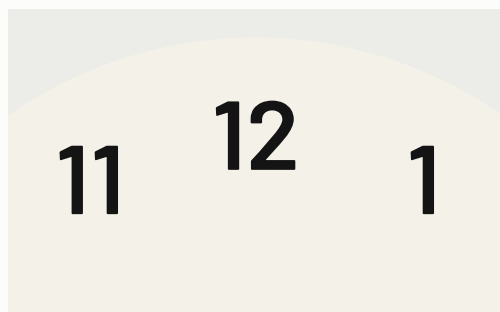
Placement

Four ways to sit.

Upright, radial, radial with auto-flip, or on path. On path sets each letter on the circle with the font's own spacing; below 3 and 9 o'clock it flips to read left to right.

Type and nudge

Each numerals ring has its own type, from any font installed on the Mac. Each numeral can then be corrected by eye.



DIN Alternate Bold



Futura Medium



Helvetica Neue Medium



Bodoni 72 Book

Type

What reaches the outlines.

Family · Style	read from macOS; no fonts folder
Instance · Axes	a variable font's named instances, and a slider per axis
Features	OpenType features under the designer's own names
Language	local forms, such as Turkish İ i
Tracking	in 1/1000 em, as in Illustrator

Nudge

Correct one numeral at a time.

Radius ±	in or out, mm
Angle ±°	round the dial
Rotate °	about its own centre
Size ×	larger or smaller

Stored per position. Reset this, or Reset all.

Typical

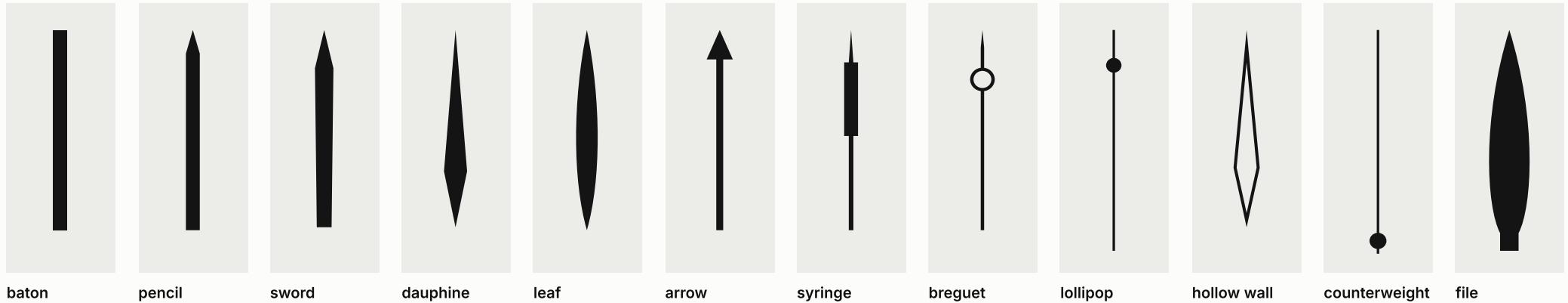
Even the weight.

Two-digit numerals look heavier than single ones. Size × 0.92 on 10, 11 and 12 evens them; Radius -0.2 pulls a wide 12 inward.

The date window uses the type of the top numerals ring.

Hands

Hour, minute and seconds each have their own shape, size and colour. Lengths are millimetres from the pivot, as movements specify them.



The diver's minute hand

Shape

Length 12.40

Width 1.50

Tail 1.80

Tip length 2.00

Counterweight Ø 0.00

Hollow wall 0.00

Feature and file

Head, ring, disc — or your own drawing.

Arrow and syringe have a head or barrel width; Breguet and lollipop a ring or disc Ø, placed at a % of the length.

A file is drawn pointing up, pivot at the artboard's centre. Fit file to length scales it to the hand's Length; Width × stretches it.

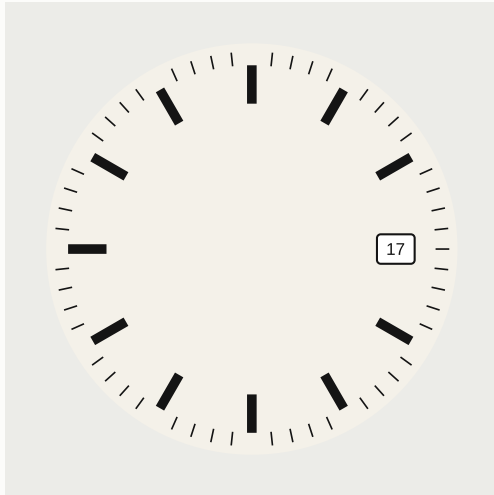
Motion

Four beats for the seconds hand.

sweep	smooth
quartz	1 step per second
6 beats/s	21,600 vph
8 beats/s	28,800 vph

Date window

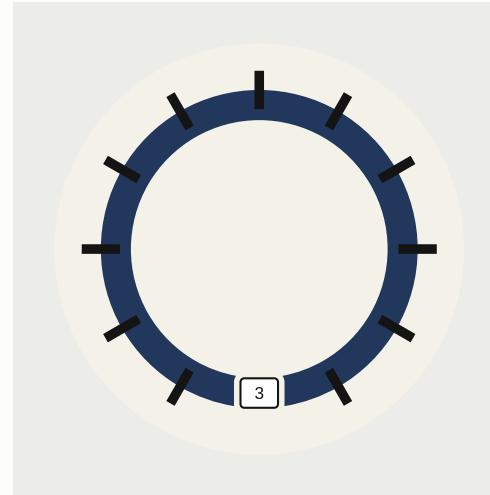
A hole in the plate that shows the date disc, with an optional printed frame. Print keeps its distance from it.



at 3
Radius 10.5 · 2.6 × 2.0 · Frame 0.15



at 4:30
Corner 0.6 · Frame 0



at 6
Print clearance 0.4 cuts the band

The window at 3

Position	3	
Radius	10.50	
Width	2.60	
Height	2.00	
Corner	0.20	
Frame	0.15	
Print clearance	0.25	

Clearance

Print keeps away from the hole.

Every ring is cut back by the frame plus Print clearance. Markers and numerals leave the date's position empty altogether, as on a real dial.

Type

The day uses the top numerals ring's font.

Shows is the day drawn on the disc; Text % of height sizes it. The frame and the day share the Print colour.

Production

The aperture becomes a cut line.

In a production mask the hole is drawn as an outline to cut or drill to.

Classics

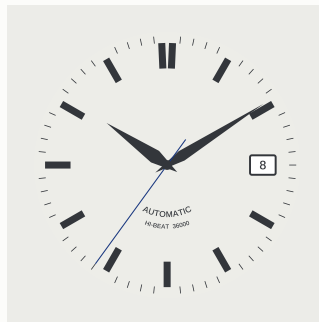
Nine dials drawn after well-known types, by eye: approximations, not measured copies. Each page prints the settings that drew the picture.



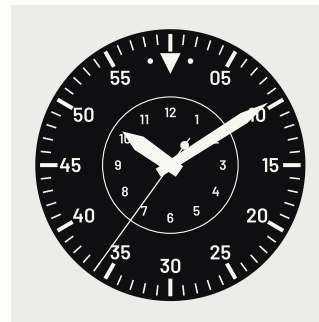
Field watch
page 15



Diver
page 16



Dress watch
page 17



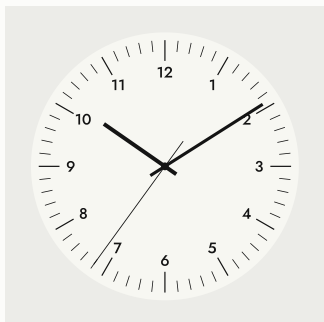
Pilot, type B
page 18



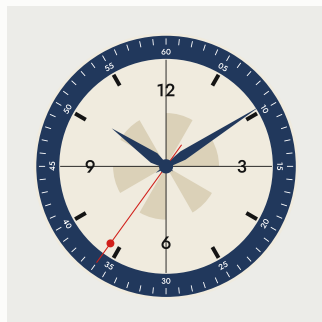
California
page 19



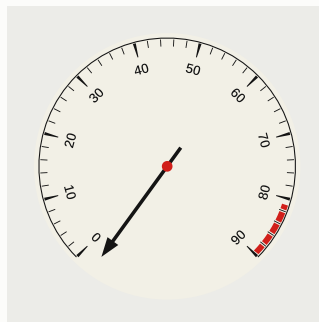
Station clock
page 20



Bauhaus
page 21



Sector dial
page 22



Gauge
page 23

Presets

Load them.

Each dial is a settings file in *docs/guide/presets*. Export → Settings → Load... opens it, ready to change.

Reading a page

Rings top first.

Only what differs from a newly added ring is printed; everything else is at its starting value.

Fonts

Named as on macOS.

DIN Alternate, Futura, Helvetica Neue, Bodoni 72. These pages were rendered off-Mac with similar open fonts, so numerals differ slightly.

Field watch

After the A-11 of the 1940s and its descendants: 12 and 24 hours, a plain minute track, nothing else.

Look for
The 24-hour ring is a second numerals ring with a Custom list. The minute ticks **skip where** the hour marks sit, so the two never overprint.

Preset
presets/field.json
Dial 30 mm ●



Rings top first, then hands. Only what differs from a new ring.

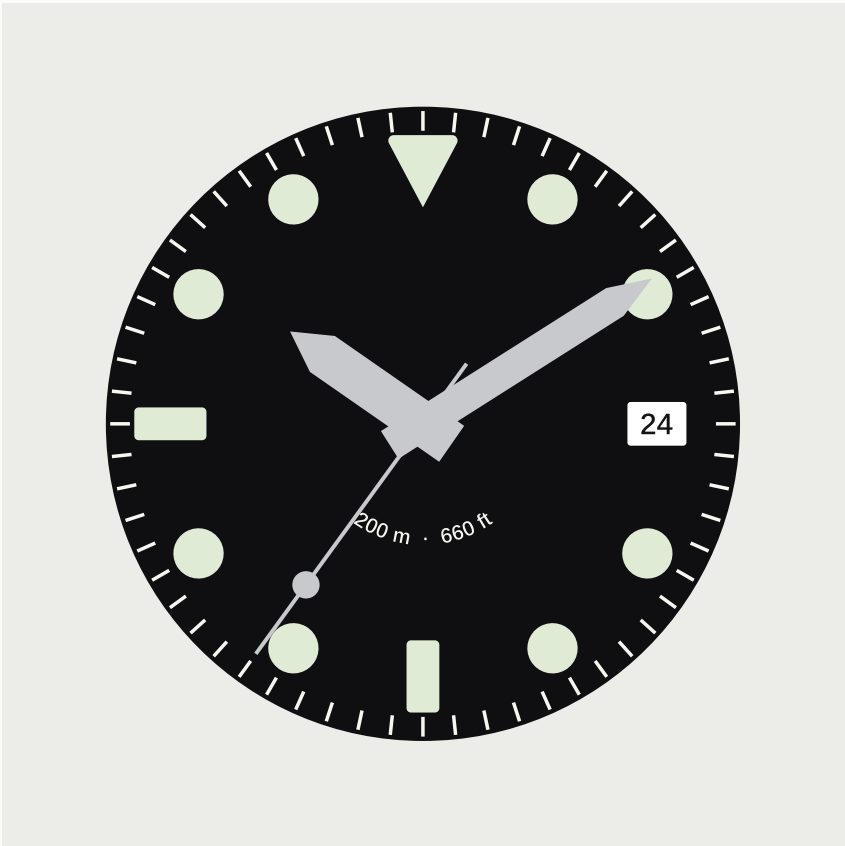
Hours numerals	Radius 10.5 · Size 3 · Font DIN Alternate Bold ●
24 hours numerals	Radius 7.3 · Labels Custom · Custom 24, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 · Size 1.35 · Font DIN Alternate Bold ●
Hour marks markers	Radius 14.4 · Length 1.7 · Width 0.42 ●
Minutes ticks	Radius 14.4 · Count 60 · Length 1 · Width 0.16 · Skip where markers above ●
Hour hand	Shape sword · Length 7.6 · Width 1.5 · Tail 1.6 · Tip length 2.4 ●
Minute hand	Shape sword · Length 12.6 · Width 1.2 · Tail 1.8 · Tip length 3 ●
Seconds hand	Shape baton · Length 13.2 · Width 0.14 · Tail 3.4 ●
Cap	Ø 1.5 ●

Diver

After the 1950s dive watch: fat dots, bars at 6 and 9, a triangle at 12, the date at 3.

Look for
Three markers rings share the hours: dots on **Every 3, skip those instead**; bars on Count 4 with the first position set to none; one triangle. The date takes 3 by itself.

Preset
presets/diver.json
Dial 29 mm ●



Rings top first, then hands. Only what differs from a new ring.

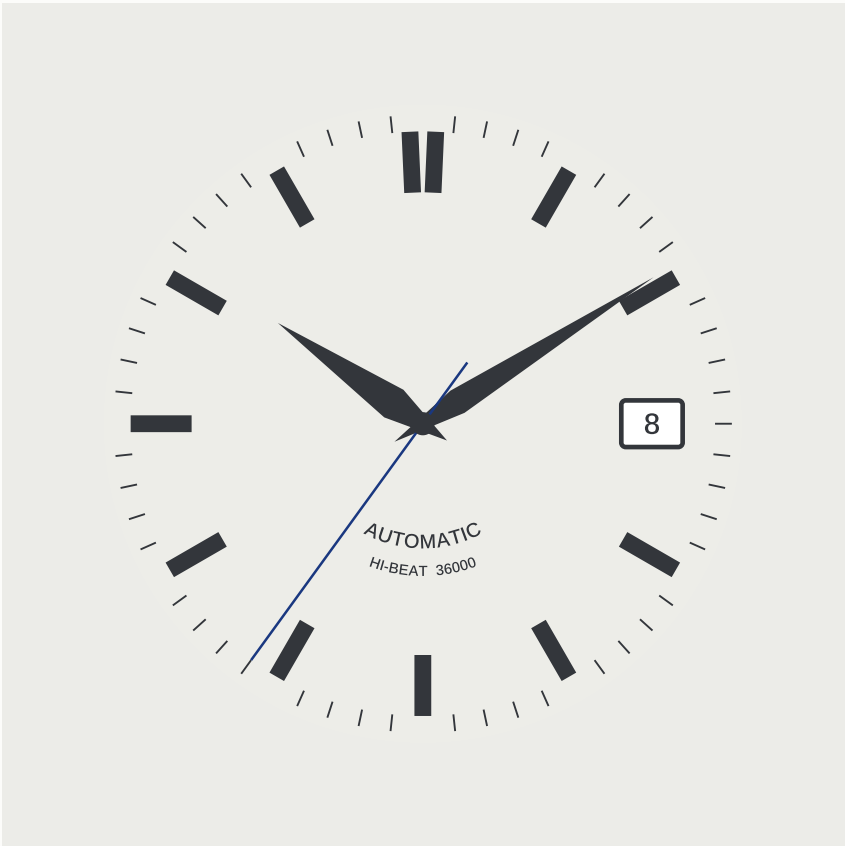
Triangle markers	Radius 13.2 · Count 1 · Shape wedge · Length 3.3 · Width 3.5 · Inner width × 0 · Round outer 0.25 ●
Bars 6 · 9 markers	Radius 13.2 · Count 4 · Length 3.3 · Width 1.5 · Round outer 0.2 · Round inner 0.2 · 12 o'clock none ●
Dots markers	Radius 13 · Every 3 · Skip those instead · Shape dot · Ø 2.3 ●
Depth numerals	Radius 5.2 · Count 1 · Start 180° · Labels Custom · Custom <i>200 m · 660 ft</i> · Placement on path · Size 0.95 · Font Helvetica Neue Medium ○
Minutes ticks	Radius 14.3 · Count 60 · Length 0.9 · Width 0.16 ○
Date window	Position 3 · Radius 10.7 · Width 2.7 · Height 2 · Corner 0.15 · Frame 0 · Print clearance 0.3 · Shows 24 · Text 68 %
Hour hand	Shape pencil · Length 7.4 · Width 2 · Tail 1.6 · Tip length 1.8 ●
Minute hand	Shape pencil · Length 12.4 · Width 1.5 · Tail 1.8 · Tip length 2 ●
Seconds hand	Shape lollipop · Length 13 · Width 0.18 · Tail 3.4 · Head · ring · disc 1.25 · at 70 % ●
Cap	Ø 1.6 ●

Dress watch

After the Japanese dress watches of the 1960s: plain bars, a double bar at 12, dauphine hands, a framed date.

Look for
The double bar is the markers ring's 12 o'clock style. Both lines of text are numerals rings with Count 1, set on path at 180°.

Preset
presets/dress.json
Dial 30.4 mm



Rings top first, then hands. Only what differs from a new ring.

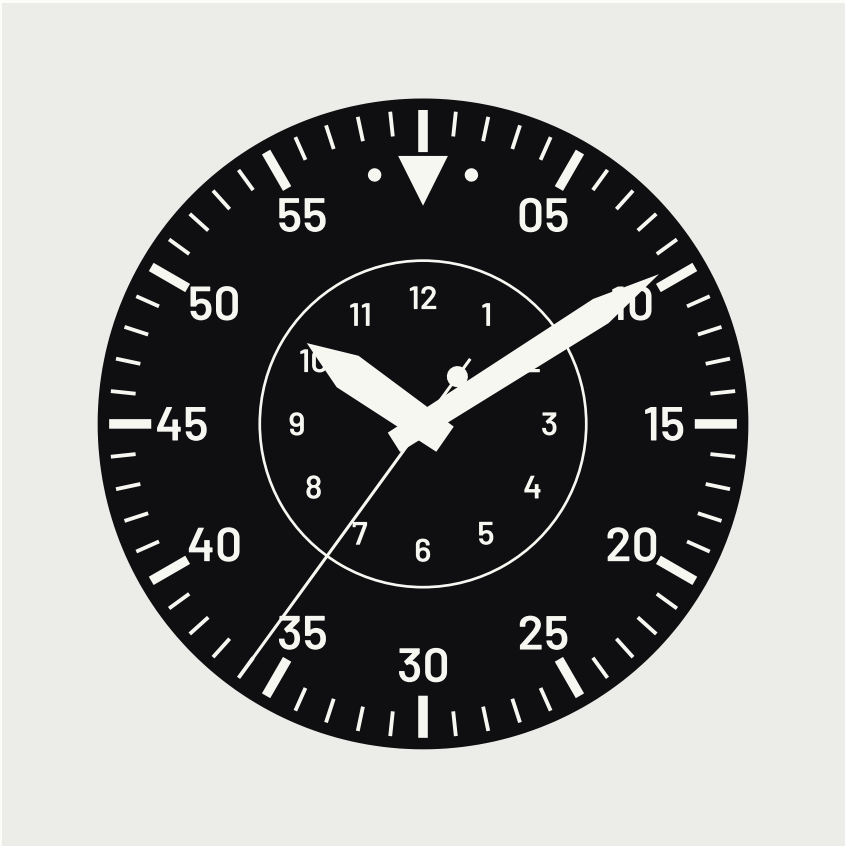
Hours markers	Radius 13.9 · Length 2.9 · Width 0.8 · 12 o'clock double · Gap 0.3
Minutes ticks	Radius 14.7 · Count 60 · Length 0.8 · Width 0.09 · Skip where markers above
Line 1 numerals	Radius 5.6 · Count 1 · Start 180° · Labels Custom · Custom AUTOMATIC · Placement on path · Size 0.95 · Font Helvetica Neue Medium
Line 2 numerals	Radius 7 · Count 1 · Start 180° · Labels Custom · Custom HI-BEAT 36000 · Placement on path · Size 0.7 · Font Helvetica Neue Medium
Date window	Position 3 · Radius 10.9 · Width 2.7 · Height 2 · Corner 0.1 · Frame 0.22 · Print clearance 0.25 · Shows 8 · Text 70 %
Hour hand	Shape dauphine · Length 8.4 · Width 1.6 · Tail 1.4 · at 20 %
Minute hand	Shape dauphine · Length 13 · Width 1.25 · Tail 1.6 · at 15 %
Seconds hand	Shape baton · Length 13.9 · Width 0.12 · Tail 3.6
Cap	Ø 1.1

Pilot, type B

After the 1940s observer's watch: minutes outside, hours on an inner ring, a triangle and two dots at 12.

Look for
The minutes are **Numbers**: first 5, step 5, two digits, over a 300° span that leaves 12 free for the triangle. The two dots are a ticks ring with Count 2 on a 22° span.

Preset
presets/pilot.json
Dial 34 mm ●



Rings top first, then hands. Only what differs from a new ring.

Minutes numerals	Radius 12.6 · Count 11 · Start 30° · Span 300° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Size 2.5 · Font DIN Alternate Bold ○
Triangle markers	Radius 14 · Count 1 · Shape wedge · Length 2.6 · Width 2.6 · Inner width × 0 ○
Two dots ticks	Radius 13.6 · Count 2 · Start -11° · Span 22° · Shape dot · Ø 0.7 ○
Five minutes markers	Radius 16.4 · Length 2.2 · Width 0.5 ○
Minute track ticks	Radius 16.4 · Count 60 · Length 1.3 · Width 0.2 · Skip where markers above ○
Hours numerals	Radius 6.6 · Size 1.7 · Font DIN Alternate Bold ○
Hour circle band	Radius 8.6 · Band width 0.14 ○
Hour hand	Shape sword · Length 7.4 · Width 2 · Tail 1.4 · Tip length 2.6 ○
Minute hand	Shape sword · Length 14.6 · Width 1.5 · Tail 1.8 · Tip length 3.6 ○
Seconds hand	Shape baton · Length 15.4 · Width 0.16 · Tail 4.2 · Counterweight Ø 1.1 ○
Cap	Ø 1.7 ○

California

Look for

Two numerals rings on one radius, each with a **Custom** list that leaves the other's positions empty — so Roman and Arabic can use different fonts and placements.

Preset

```
presets/california.json
```

Dial 30 mm ●



Rings top first, then hands. Only what differs from a new ring.

Roman numerals	Radius 9.6 · Labels Custom · Custom , <i>l</i> , <i>ll</i> , <i>lll</i> , <i>llll</i> , <i>lllll</i> , <i>llllll</i> , <i>lllllll</i> , <i>llllllll</i> , <i>lllllllll</i> · Placement radial · Size 3.3 · Font Bodoni 72 Book 
Arabic numerals	Radius 9.6 · Labels Custom · Custom , <i>l</i> , <i>ll</i> , <i>lll</i> , <i>llll</i> , <i>lllll</i> , <i>llllll</i> , <i>lllllll</i> , <i>llllllll</i> , <i>lllllllll</i> · Placement upright · Size 3.2 · Font DIN Alternate Bold 
Bars + triangle markers	Radius 11.4 · Count 4 · Length 3.4 · Width 1 · 12 o'clock triangle · Scale × 1.25 
Minutes ticks	Radius 14.2 · Count 60 · Length 1 · Width 0.14 
Outer rail band	Radius 14.2 · Band width 0.12 
Inner rail band	Radius 13.32 · Band width 0.12 
Hour hand	Shape pencil · Length 7.2 · Width 1.5 · Tail 1.4 · Tip length 1.6 
Minute hand	Shape pencil · Length 11.8 · Width 1.2 · Tail 1.6 · Tip length 1.8 
Seconds hand	Shape baton · Length 12.6 · Width 0.13 · Tail 3.2 · Counterweight Ø 0.9 
Cap	Ø 1.4 

Station clock

After the Swiss railway clock of 1944: heavy bars, and a red seconds hand like a dispatcher's baton.

Look for
Two rings. Everything is in the proportions:
bars 3.7×1.3 , ticks 1.3×0.45 , and a
lollipop with its disc at 100 % of the length.

Preset
presets/railway.json
Dial 30 mm ○



Rings top first, then hands. Only what differs from a new ring.

Hours markers	Radius 14.3 · Length 3.7 · Width 1.3 ●
Minutes ticks	Radius 14.3 · Count 60 · Length 1.3 · Width 0.45 · Skip where markers above ●
Hour hand	Shape baton · Length 8.8 · Width 1.7 · Tail 2.4 ●
Minute hand	Shape baton · Length 13.6 · Width 1.3 · Tail 2.8 ●
Seconds hand	Shape lollipop · Length 9.8 · Width 0.34 · Tail 3.6 · Head · ring · disc 2.7 · at 100 % ●
Cap	Ø 1 ●

12.7

Bauhaus

After the German wall-clock school of the 1950s–60s: hairline marks, small numerals, thin hands.

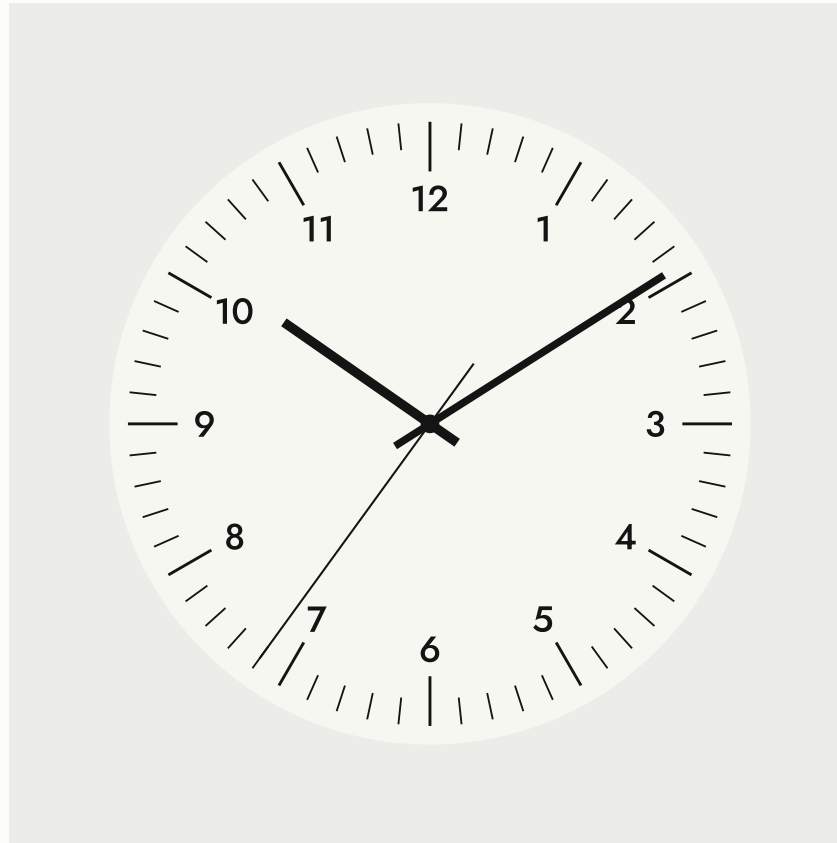
Look for

Line weights carry it: 0.14 for the hours, 0.09 for the minutes, hands under half a millimetre.

Preset

presets/bauhaus.json

Dial 31 mm ○



Rings top first, then hands. Only what differs from a new ring.

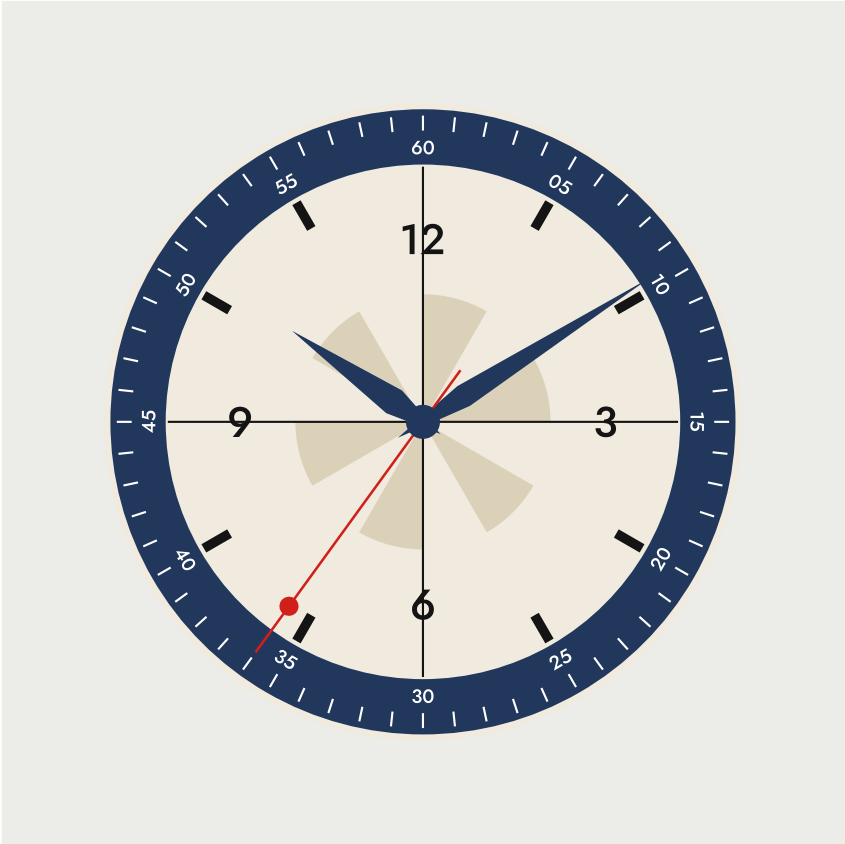
Hours numerals	Radius 10.9 · Size 1.75 · Font Futura Medium ●
Hour lines markers	Radius 14.6 · Length 2.4 · Width 0.14 ●
Minutes ticks	Radius 14.6 · Count 60 · Length 1.3 · Width 0.09 · Skip where markers above ●
Hour hand	Shape baton · Length 8.6 · Width 0.46 · Tail 1.6 ●
Minute hand	Shape baton · Length 13.4 · Width 0.36 · Tail 2 ●
Seconds hand	Shape baton · Length 14 · Width 0.1 · Tail 3.6 ●
Cap	Ø 0.9 ●

Sector dial

After the 1930s scientific dials: a chapter ring, a crosshair, zones of colour.

Look for
Bands do the colour: a chapter ring 2.6 wide, and pies made by Band width = Radius with Every 2. The crosshair is a ticks ring, Count 4, Length 12.

Preset
presets/sector.json
Dial 30 mm



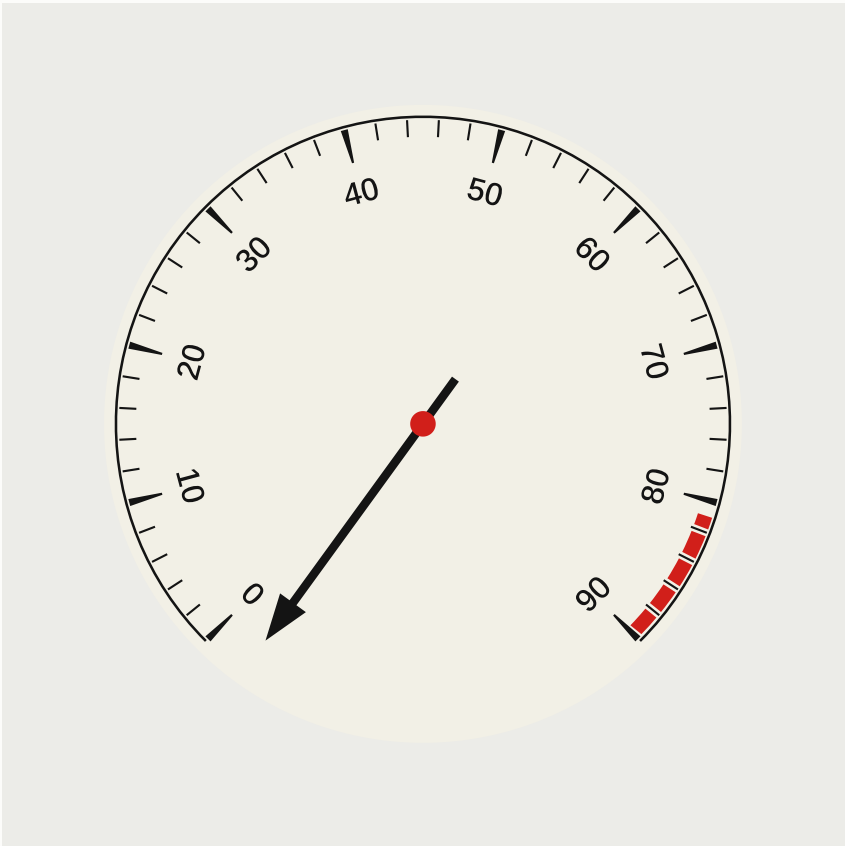
Rings top first, then hands. Only what differs from a new ring.	
Hours numerals	Radius 8.6 · Every 3 · Size 2 · Font Futura Medium ●
Hour bars markers	Radius 11.9 · Every 3 · Skip those instead · Length 1.4 · Width 0.45 ●
Minutes ticks	Radius 14.4 · Count 60 · Length 0.7 · Width 0.1 ○
Minute numerals	Radius 12.9 · Start 30° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Placement radial, auto-flip · Size 0.9 · Font Futura Medium ○
Chapter band	Radius 14.7 · Band width 2.6 ●
Crosshair ticks	Radius 12 · Count 4 · Length 12 · Width 0.1 ●
Sectors band	Radius 6 · Count 12 · Every 2 · Band width 6 ○
Hour hand	Shape dauphine · Length 7.5 · Width 1.3 · Tail 1 · at 22 % ●
Minute hand	Shape dauphine · Length 12.6 · Width 1.1 · Tail 1.4 · at 18 % ●
Seconds hand	Shape lollipop · Length 13.4 · Width 0.12 · Tail 3 · Head · ring · disc 0.9 · at 80 % ●
Cap	Ø 1.6 ●

Gauge

A 270° scale with a red zone: positions over a partial span mark both ends.

Look for
Start -135°, Span 270°. A partial span marks both ends, so Count 10 gives 0 to 90. The red zone **knocks out** everything above it with a 0.12 clearance.

Preset
presets/gauge.json
Dial 30 mm ○



Rings top first, then hands. Only what differs from a new ring.

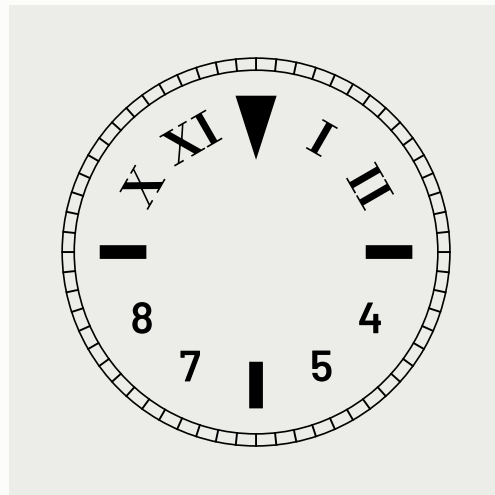
Scale numerals	Radius 11.3 · Count 10 · Start -135° · Span 270° · Labels Numbers · First number 0 · Step 10 · Placement radial, auto-flip · Size 1.5 · Font Helvetica Neue Medium ●
Scale marks markers	Radius 14.3 · Count 10 · Start -135° · Span 270° · Shape wedge · Length 1.6 · Width 0.34 · Inner width × 0.3 ●
Scale ticks ticks	Radius 14.3 · Count 46 · Start -135° · Span 270° · Length 0.8 · Width 0.1 · Skip where markers above ●
Rail band	Radius 14.5 · Count 1 · Start -135° · Span 270° · Band width 0.12 ●
Red zone band	Radius 14.5 · Count 1 · Start 108° · Span 27° · Band width 0.9 · Knock out everything above · Clearance 0.12 ●
Minute hand	Shape arrow · Length 12.6 · Width 0.4 · Tail 2.6 · Tip length 2.2 · Head · ring · disc 1.5 ●
Cap	Ø 1.2 ●

Files in and out

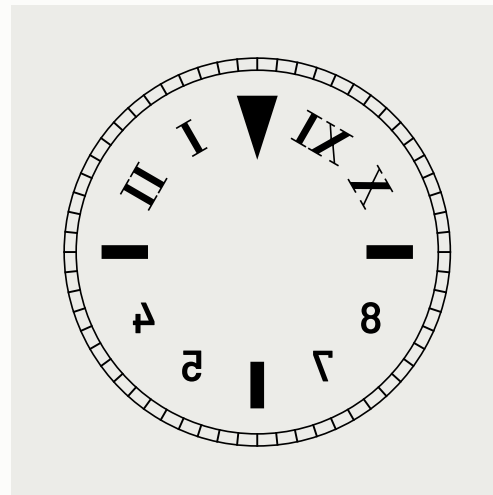
Vectors at true size, a one-colour mask for production, and your own shapes coming in.



design
PDF · SVG · PNG



production
one colour, no plate, no hands



production + mirror
for toner transfer

Export

☒ Production: mono print mask
☒ Mirror: toner transfer
☐ Live text in PDF · SVG (fonts needed)

PNG dpi

Clip seconds

Frames per s

PDF · SVG · PNG

True size.

Vectors for Illustrator, print, the press, laser: a 30 mm dial opens 30 mm wide. PNG at your dpi, for film: 30 mm at 600 dpi is 709 pixels.

MP4 · GIF

The hands moving.

From the set time, for Clip seconds at Frames per s, in the chosen beat.

Live text

Off by default.

Keeps numerals editable in PDF and SVG. The fonts are then needed, and on-path text becomes one object per letter.

In: SVG · PDF · AI

Your own markers and hands.

Draw pointing up; the artboard's centre is the anchor. Black is ink, white cuts holes, strokes become outlines. Outline text first; images and gradients are skipped.

Recipes and help

Short constructions to build on, each with a dial that uses it. Then what to check when something looks wrong.

Track

Railroad

Ticks, Count 60. Two bands 0.12 wide: one at the ticks' Radius, one at Radius – tick Length + 0.12. Page 19.

Track

Fifths of a second

Ticks Count 60 above ticks Count 300. The 300 Skip where: all marks above.

Hours

24-hour ring

Numerals, Labels Custom: 24, 13, 14 ... 23, at a smaller Radius and Size. Page 15.

Hours

Two fonts, one circle

Two numerals rings on one Radius; each Custom list leaves the other's positions empty. Page 19.

Minutes

05 to 60

Numerals, Labels Numbers, First number 5, Step 5, Digits 2, Start 30°. Page 22.

Text

A wordmark

Numerals, Count 1, Start 0° or 180°, Labels Custom, Placement on path. Page 17.

Colour

A gap round numerals

A band below the numerals; Knock out: markers & numerals above; Clearance 0.3.

Colour

Sector dial

A band, Count 12, Every 2, Band width equal to Radius. Page 22.

A ring is gone

Three places to look

Its checkbox in the list; its Every and Offset; and whether it skips positions that a ring above takes.

Diameter

Nothing else moves

Radii and hand lengths are millimetres from the centre, as on a real movement. Change them ring by ring.

Slow

Check the log

It should say “shape combining: skia-pathops”. If not, install it: page 2.

Start over

Defaults

Export → Settings → Defaults. The last session is in *~/Library/Application Support/DialTool*.

When something is wrong

Read the log, compare the previews, send both.

- 1 Open Export and read the log. At launch it says “preview: canvas” and “all controls connected”.
- 2 If the preview looks wrong, switch Canvas to PDF and compare.
- 3 Send a screenshot and the log. If the window doesn't build at all, run *tools/dial_probe.py* in DrawBot and send its report.

Where things live

One repository.

Code and docs

github.com/duztman/dial-tool

Words

docs/glossary.md

Presets

docs/guide/presets

This guide

docs/guide/guide.html, built by *tools/make_guide.py*; its rules are in *docs/design.md*